

Report for Assignment 2

(Fairness and Bias in Predictive Modeling)

1. Introduction

The primary objective of this project is to predict whether a student will graduate, remain enrolled, or drop out, based on academic, demographic, and socioeconomic factors. This assignment focuses on fairness-aware machine learning. This means, in addition to high accuracy, the system should ensure that predictions are not biased against specific groups, such as gender or nationality.

2. Dataset Summary

We used the dataset "Predict students' dropout and academic success" from the UCI Repository.

Number of records: 4,424 students.

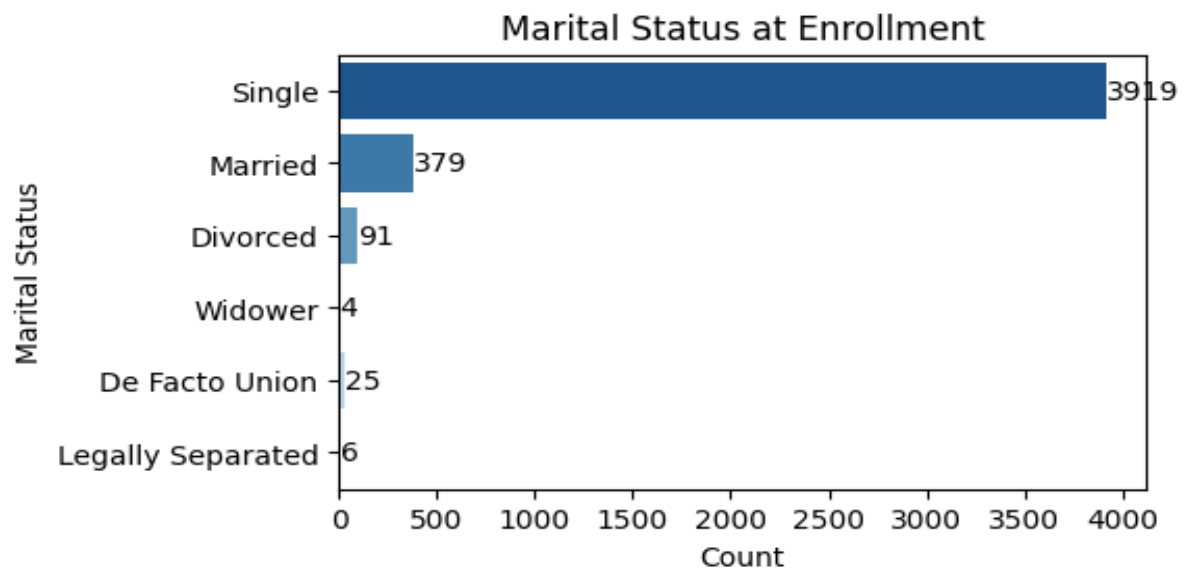
Number of features: 36 input features and 1 target label.

Target variable:

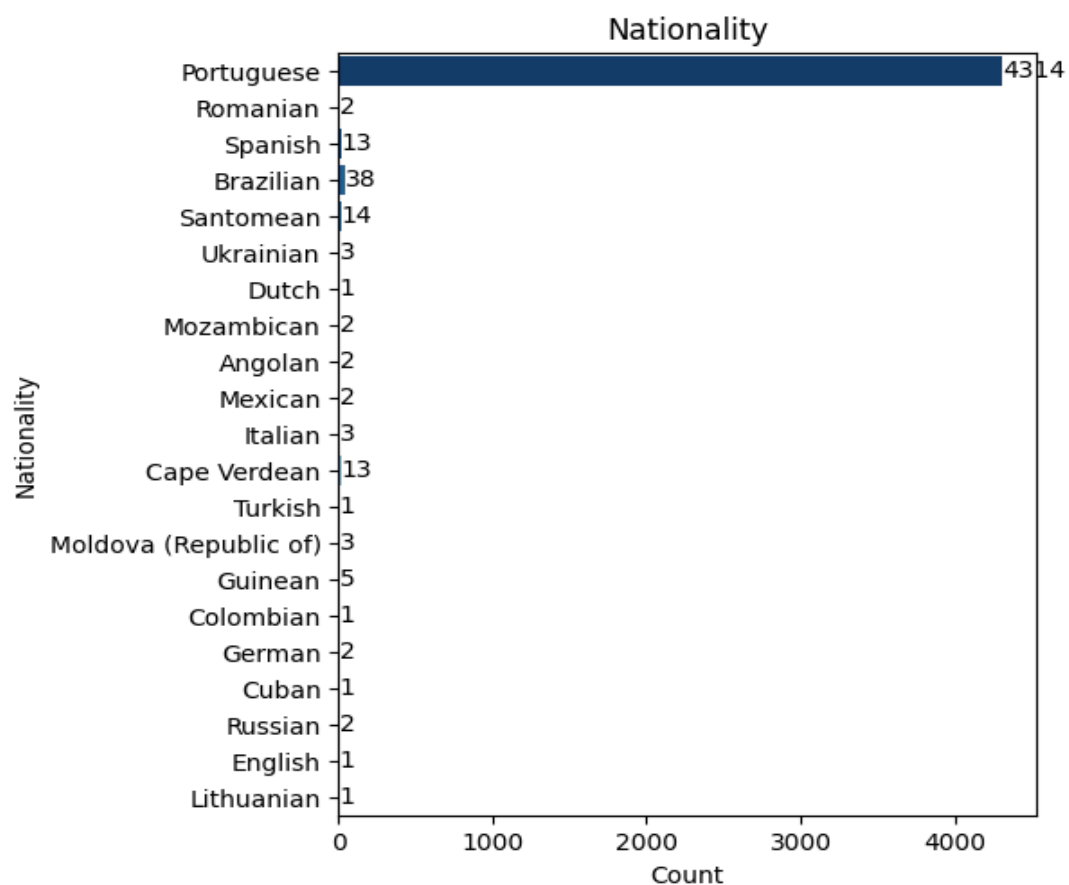
- Dropout (student left the course),
- Enrolled (student is continuing),
- Graduate (student completed the program).

3. Data exploration with Feature Groups

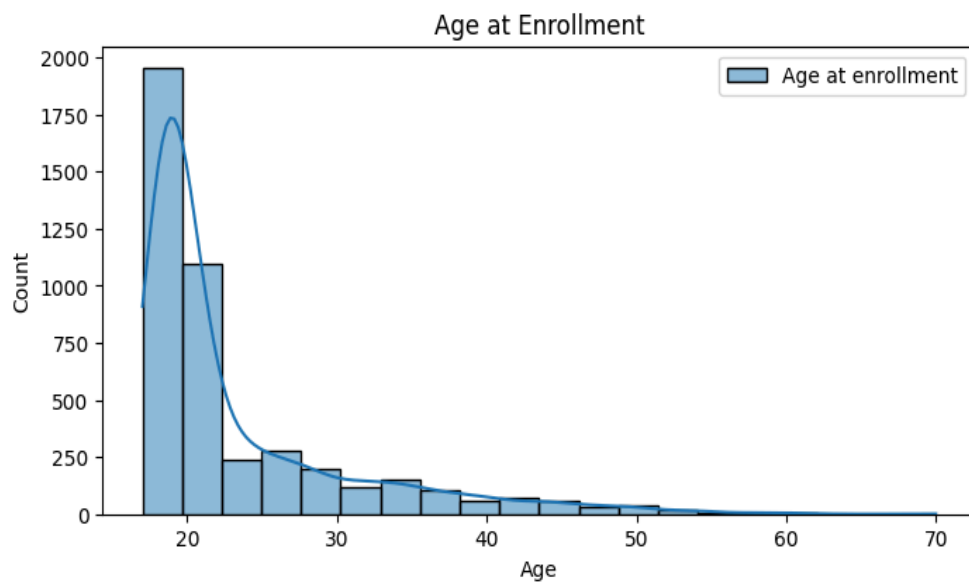
- Demographic: Gender, age, marital status, nationality.



The majority of students are single, followed by married, and then divorced. Only a small proportion fall under categories such as de facto union, legally separated, or widowed.

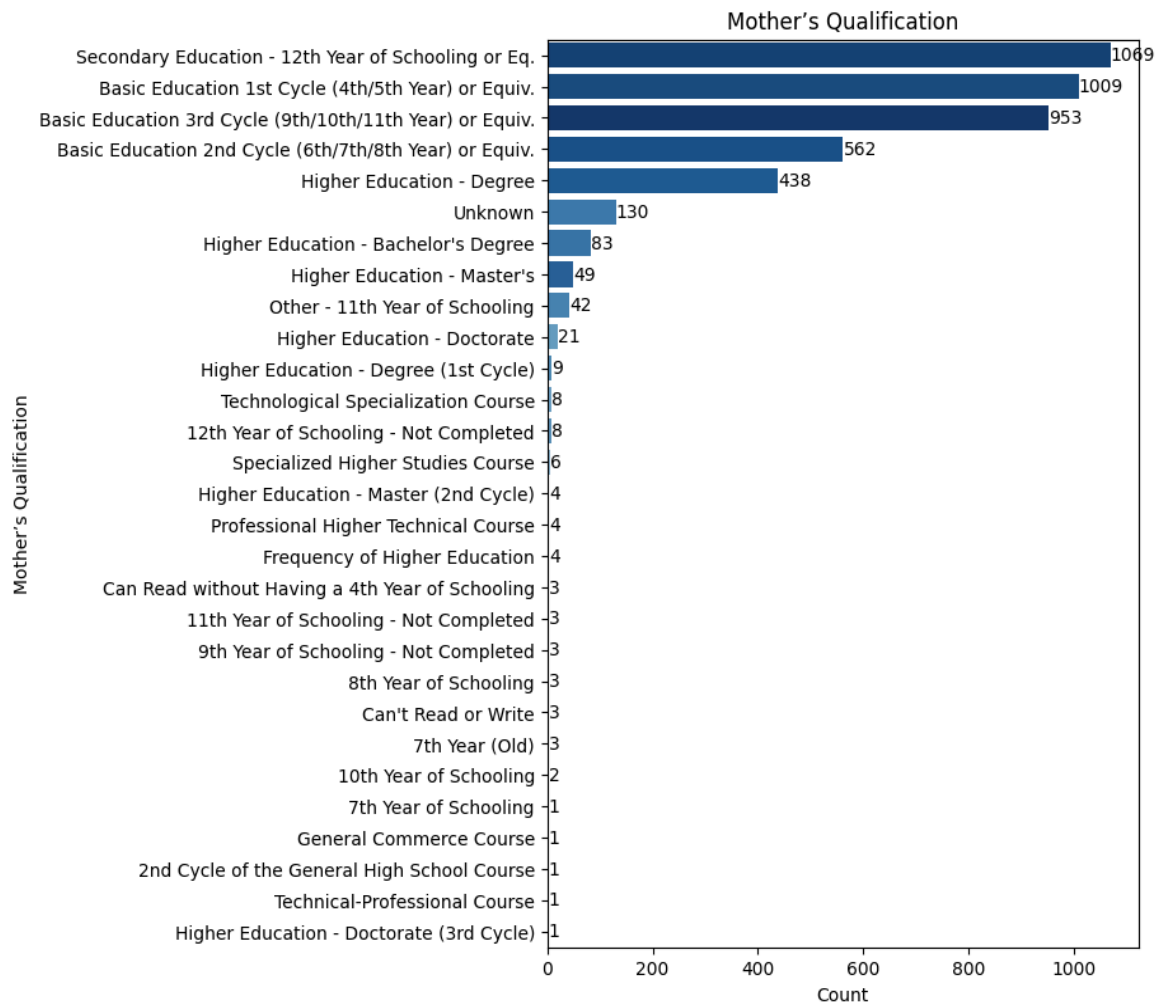


Most of the students are from Portugal, followed by Brazil

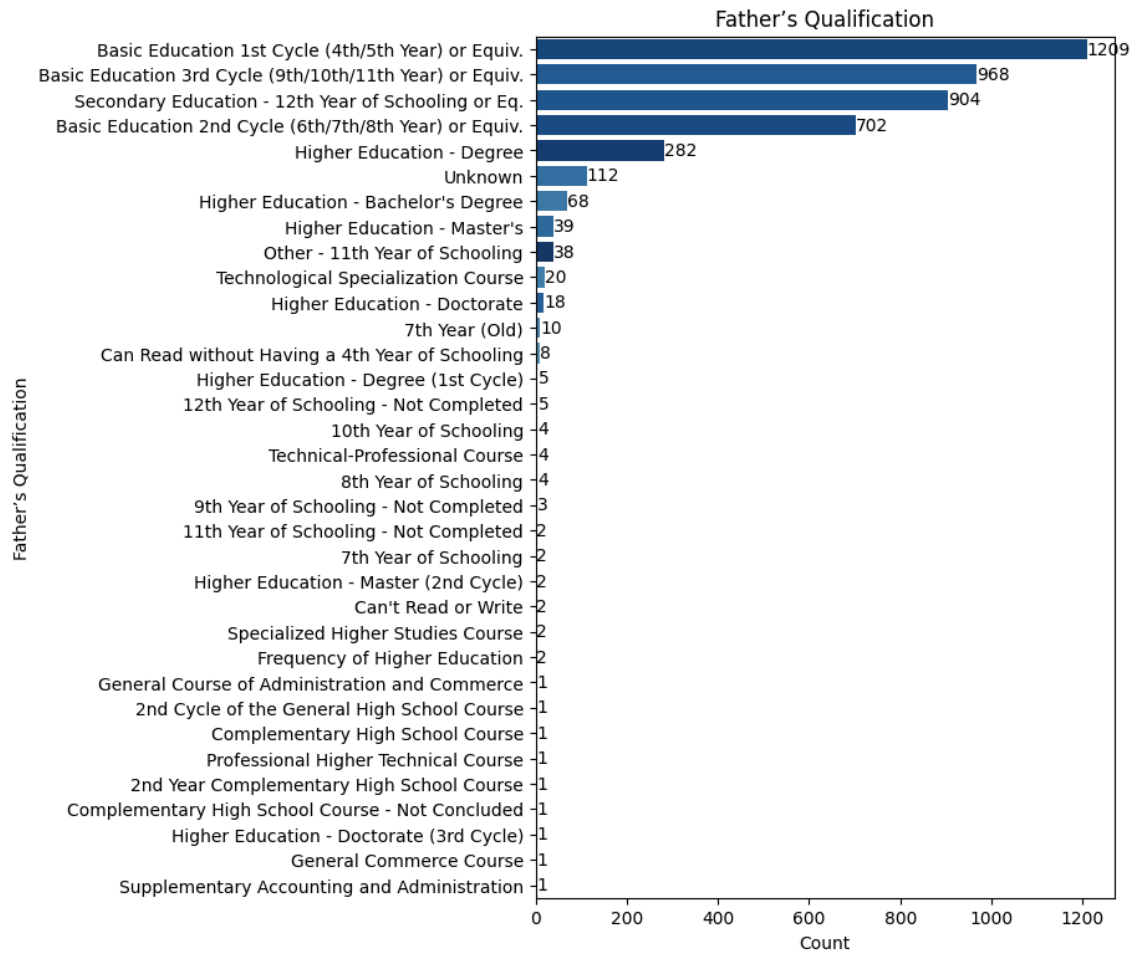


Most Students are young, while very few are above the age of 30 and pursue studies.

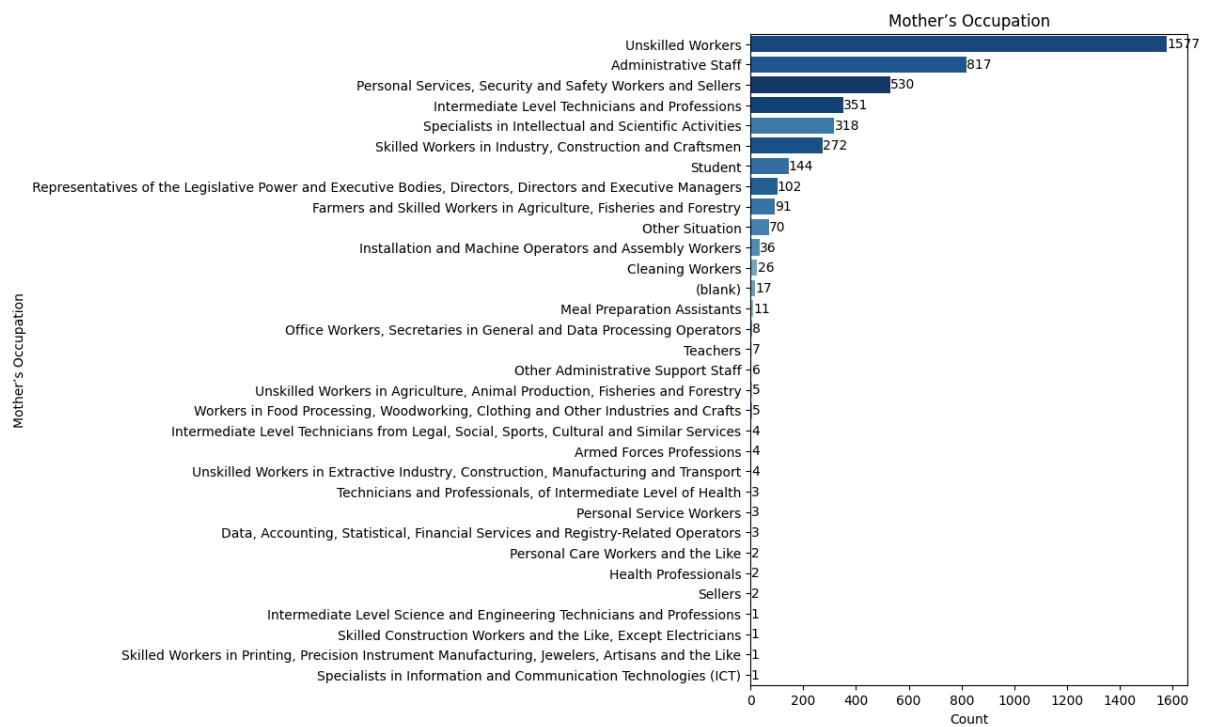
- Socio-economic: Parents' education, occupation, scholarship holder, debtor status, and tuition fee up to date.



Most of the mothers' qualifications are Basic Education, followed by Secondary Education. Very few have a Degree.

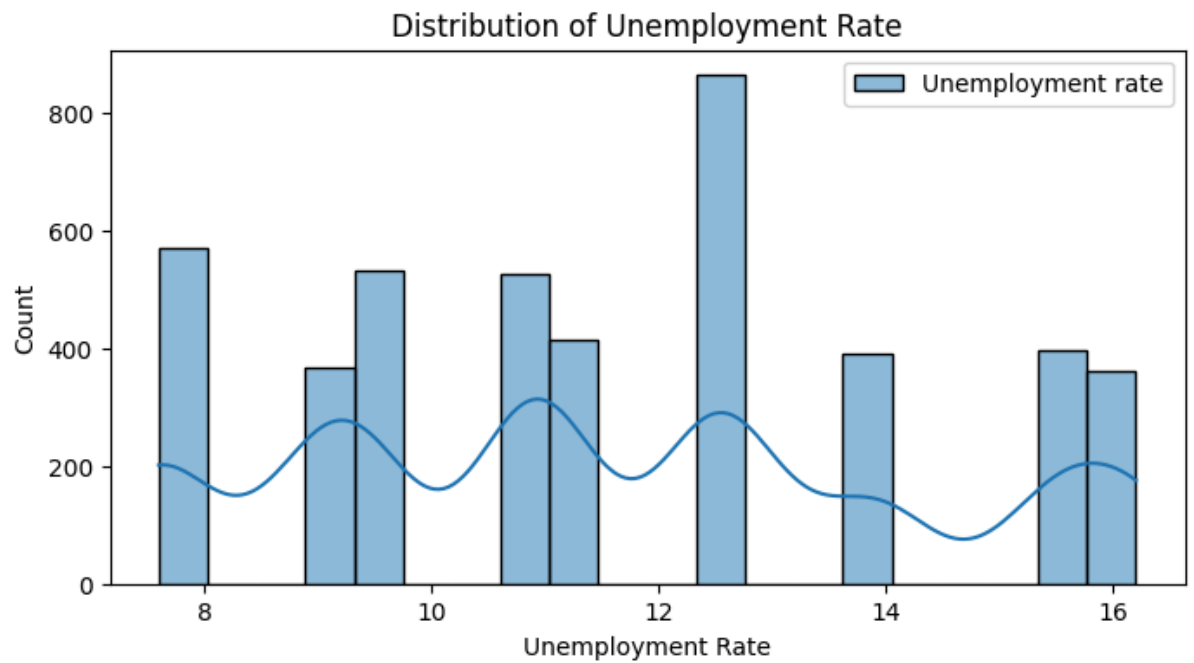


Most of the Father's qualifications are Basic Education, followed by Secondary Education. Very few have a Degree.

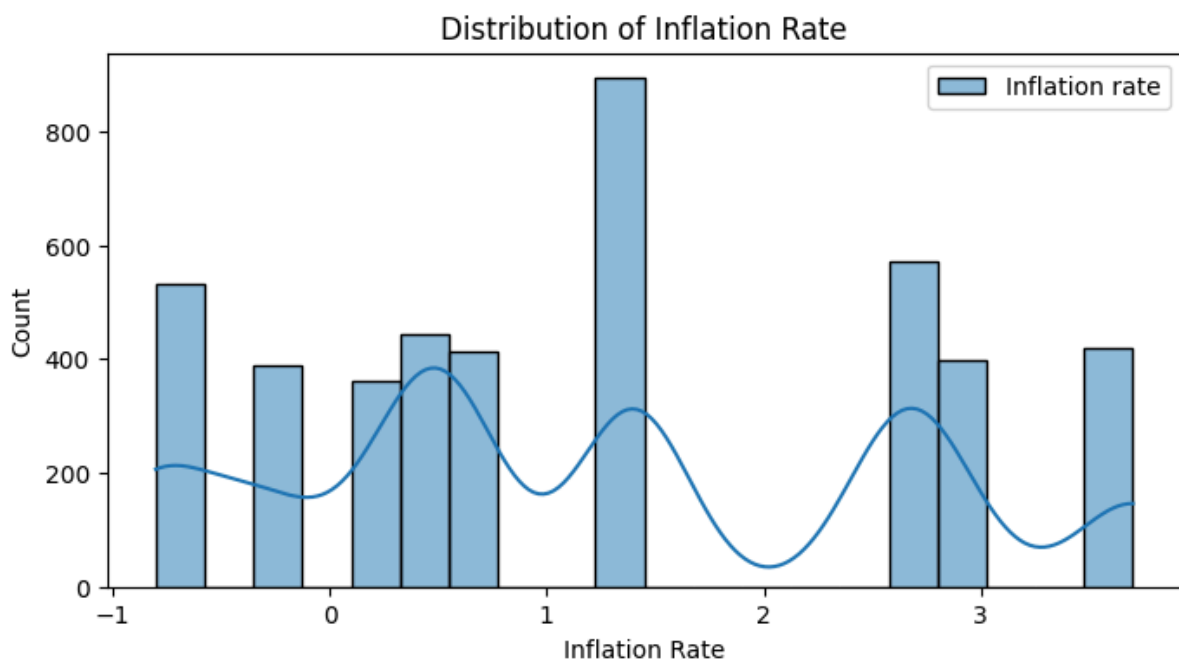


Mostly Mothers are “Unskilled workers”, followed by “Administrative Staff” and “Personal Services, Security and Safety Workers and Sellers”

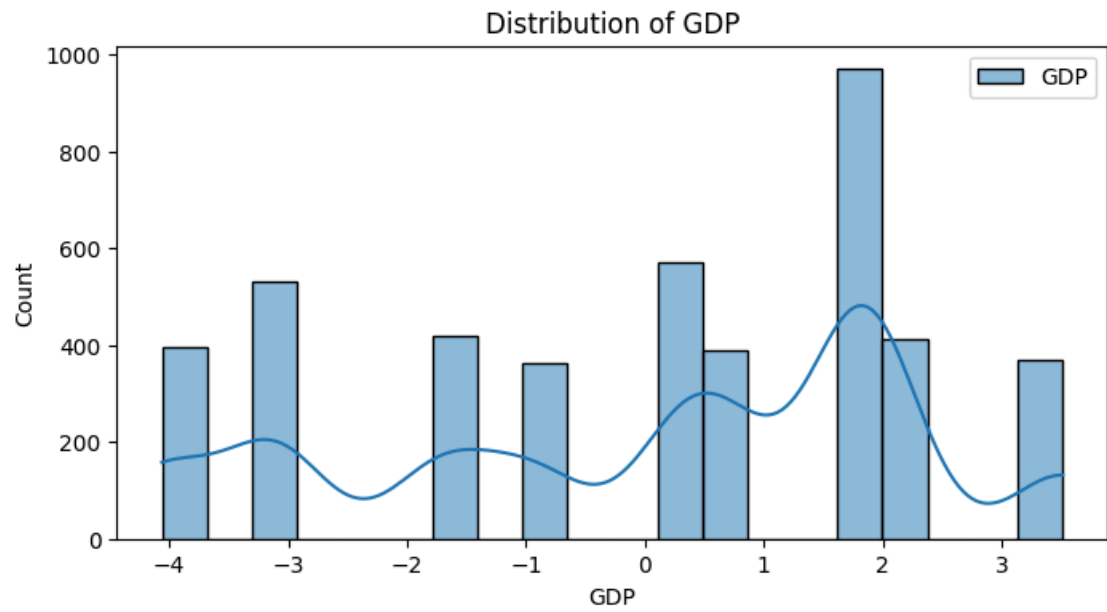
- Economic: Country-level stats like GDP, unemployment rate, and inflation.



The unemployment rate shows moderate variation, but its weak correlation with academic outcomes suggests limited predictive power

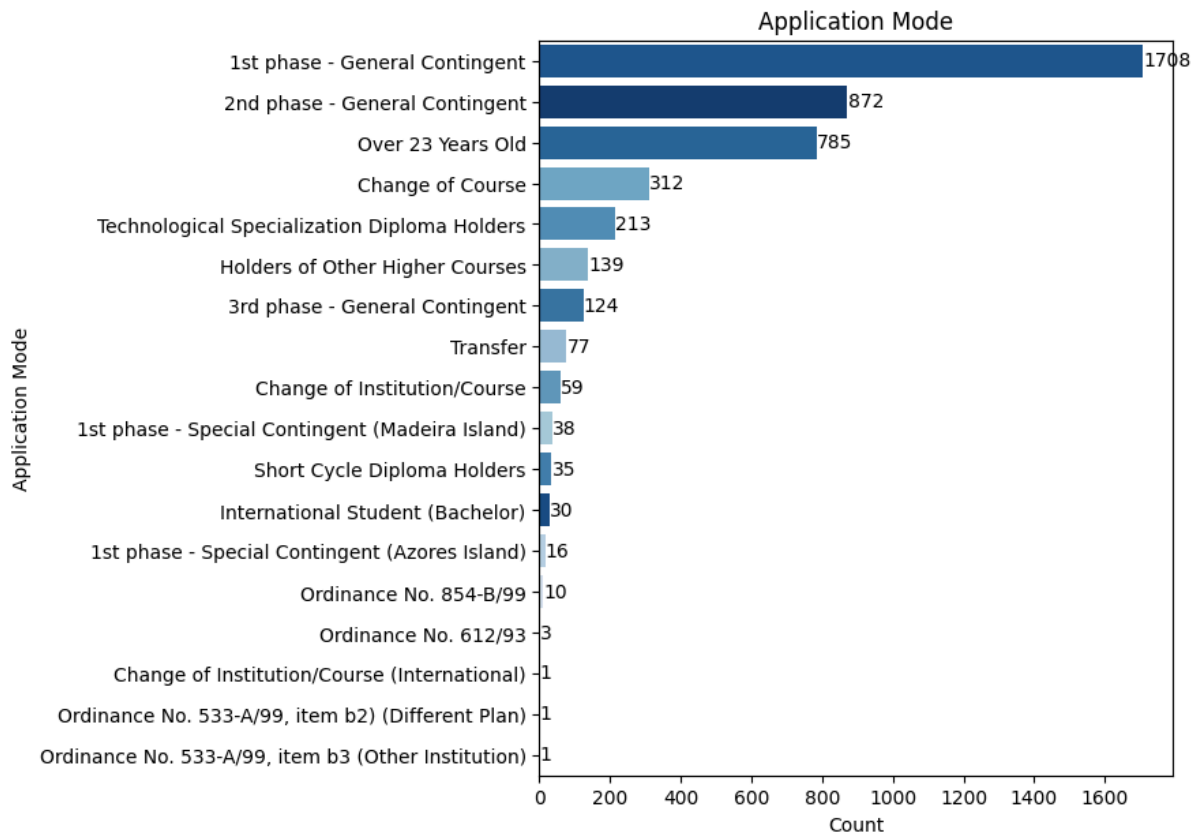


The inflation rate fluctuates across values, yet it has minimal direct influence on individual student success or dropout

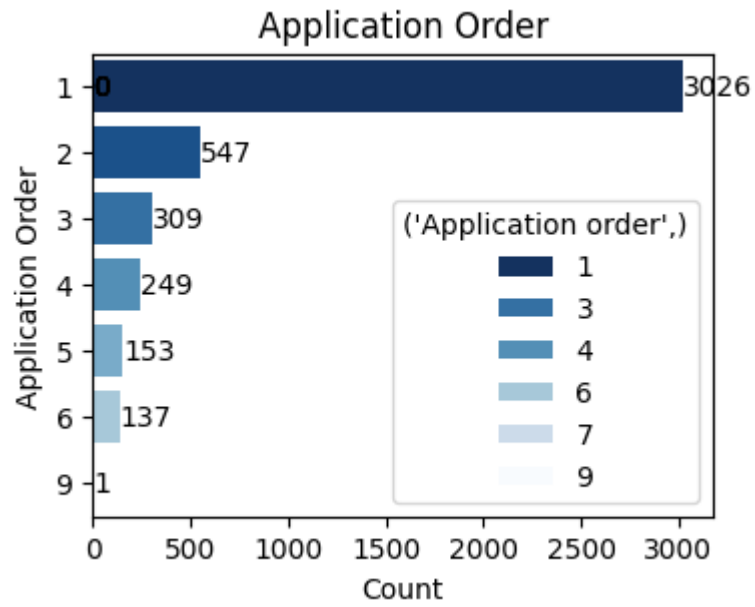


GDP values are spread across positive and negative ranges, though they exhibit little connection to student-level academic performance

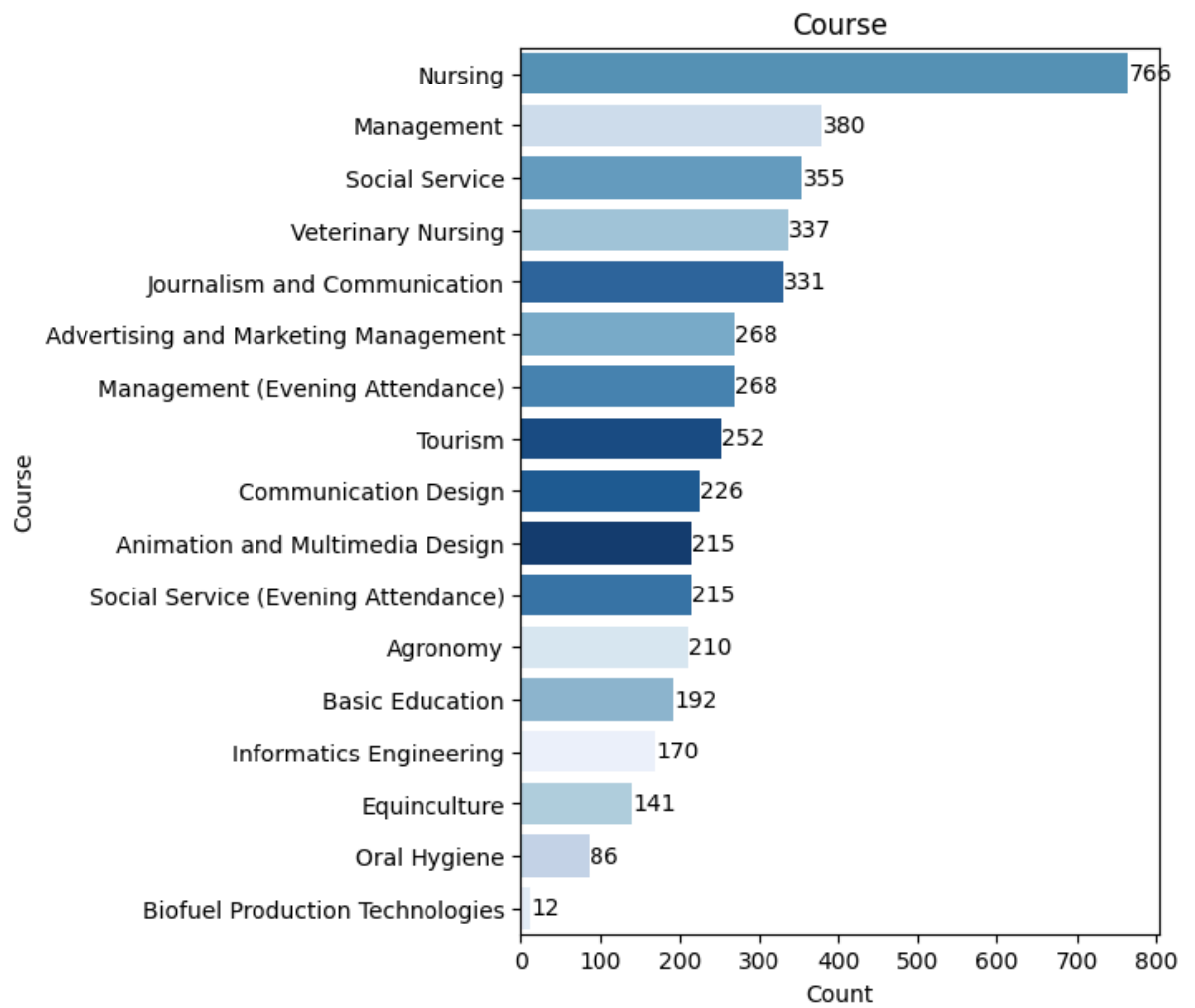
- Academic: Admission grade, course, number of curricular units enrolled and approved in the 1st and 2nd semester, average grades, evaluations.

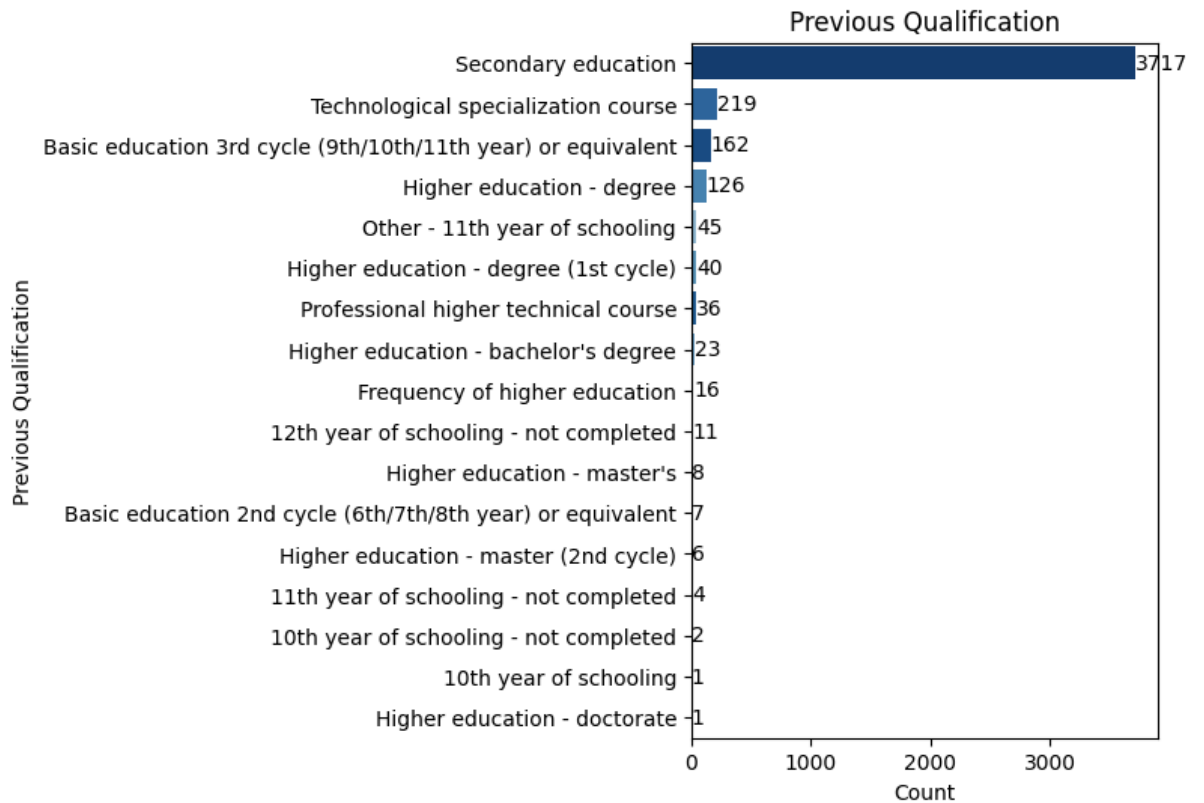


Most students enter through the 1st and 2nd phase general contingents, while other modes like transfers or international admissions are far less common

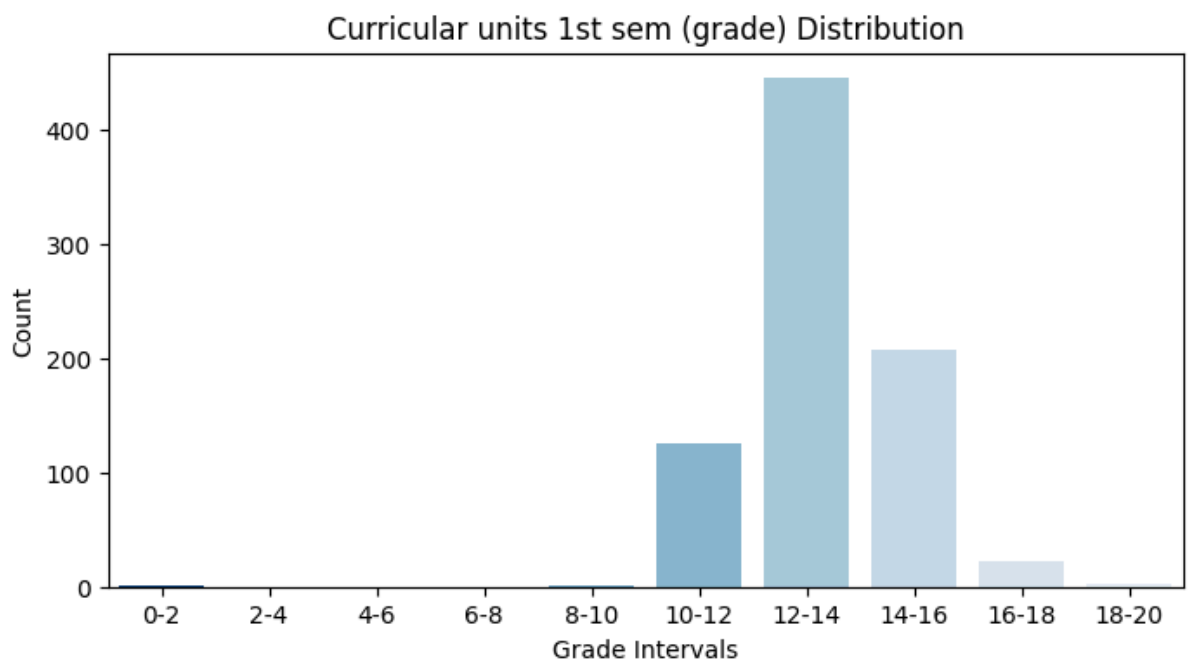


The majority of students are admitted to their first-choice course, with far fewer placed in lower-ranked options

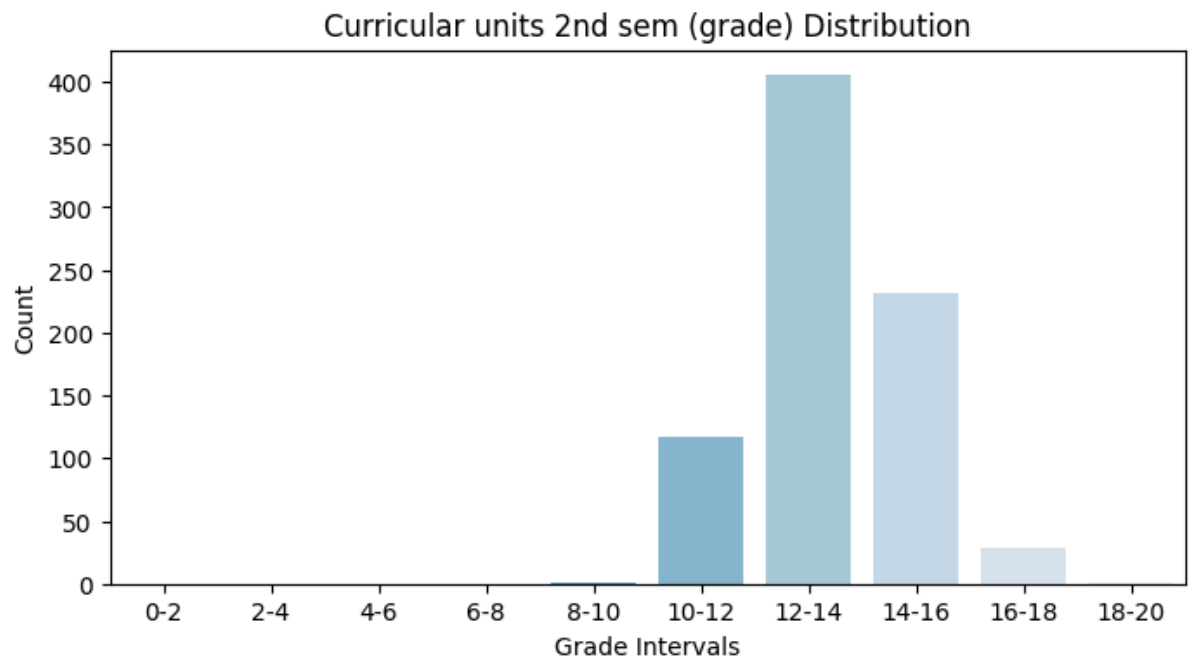




Most students come with a secondary education background, with smaller proportions from technological specialization or higher education degrees



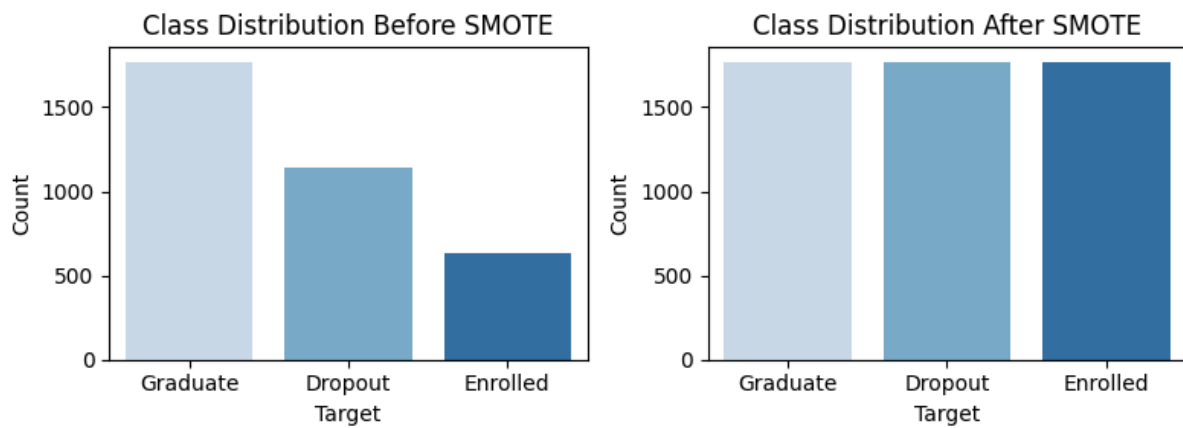
Most students score between 12–14 in the first semester, with fewer achieving higher or lower grade ranges



Second-semester grades also cluster around 12–14, showing consistency in academic performance across semesters

4. Imbalance Data Handling:

The dataset is imbalanced, with most students either graduating or enrolling, while fewer drop out.



5. Bias & Fairness Analysis

Potential Bias Sources

Gender

- Sensitive attribute
- Models can favor one gender over another (e.g., systematically predicting higher dropout risk for females vs. males).

International (Domestic vs. International student)

- Sensitive attribute.
- May reflect cultural/linguistic bias or systemic differences in support.

Scholarship holder

- Socioeconomic fairness attribute.
- Bias can emerge if students without scholarships are predicted to have higher dropout risk **solely due to financial status**.

Age at enrollment (context-dependent)

- Not always legally “protected,” but older students may face systemic disadvantages.
- Could reflect indirect **age bias** in predictions.

Nationality (optional, depending on use case)

- Sensitive attribute if used, could introduce bias against specific countries/regions.
- In practice, it is often dropped because it adds little predictive value but strong potential for discrimination.

Potential Fairness Sources

Fairness analysis focuses on features that relate to **demographic or social groups** (e.g., gender, race, nationality, financial aid).

In the dataset:

Gender: Male vs. Female is directly tied to equal opportunity concerns.

International: Domestic vs. International students linked to potential discrimination in admission/academic support.

Scholarship Students with vs. without financial support → socioeconomic fairness concern

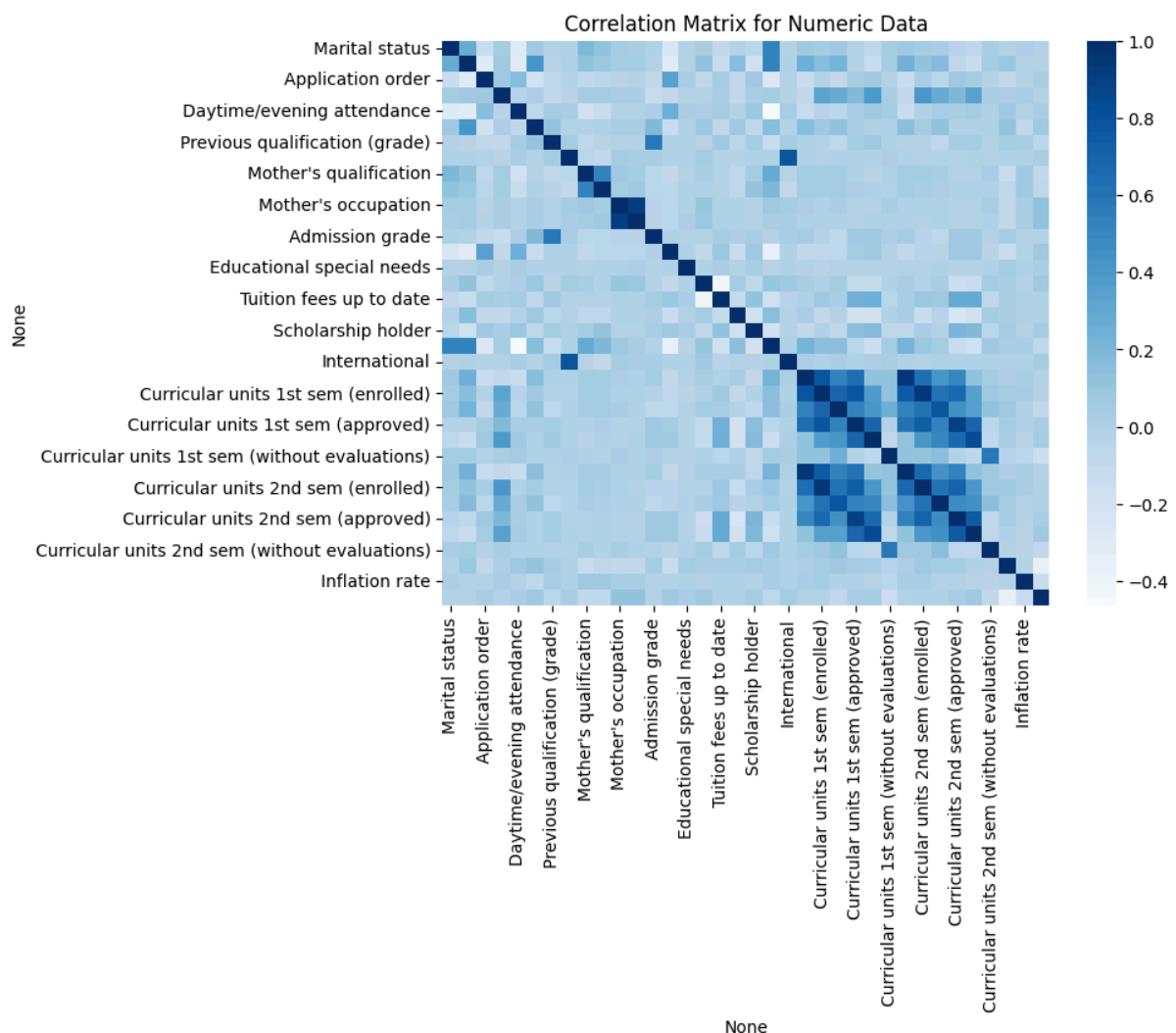
6. Corrective Measures

We handle bias & fairness, we use the Pearson correlation coefficient, domain knowledge, and Interpretability.

Feature Importance and Correlation Analysis

In this dataset, not all features contribute equally to predicting student dropout risk. We used a three-step method to identify the most relevant features:

- **Correlation Analysis.** These highlighted features had a strong connection with the target variable.
- **Domain Knowledge.** We evaluated feature relevance based on educational research and prior understanding.
- **Interpretability.** We ensured the selected features are easy to understand and apply in real-world situations.



Key Findings from Correlation Analysis

- Strong Correlations

- Academic features, such as curricular units (enrolled, credited, approved, and evaluated) and grades, showed strong positive correlations with one another.
- For example, students who enroll in more units generally complete or approve more and achieve higher evaluation activity.
- Implication: This indicates feature redundancy, suggesting the need for dimensionality reduction or careful selection.

- Moderate Correlations

- Admission grades correlated with overall curricular performance, showing a consistent academic pattern.
- Tuition fee status correlated with debtor status, reflecting financial consistency.

- Weak Correlations

- Demographic features like marital status, father's occupation, nationality, and gender showed weak correlations with academic outcomes.
- Implication: These features are less predictive compared to academic and financial features.

- Macroeconomic Variables

- Unemployment rate, inflation, and GDP showed very weak correlations, indicating limited direct impact on student-level academic persistence.

Feature Groups

- Academic Performance (Strong Predictors)

- Admission grade: Higher scores are linked with persistence.
- Curricular units (enrolled, credited, approved, evaluations, grades): Indicators of engagement and early success. Students who fail to complete early courses are more likely to drop out.

- Financial and Administrative Indicators

- Debtor status: Increases dropout risk.
- Tuition fee status: Strong evidence of financial stability.
- Scholarship holder: Lowers dropout risk by easing financial stress.

- Demographic and Background Factors

- Age: Older students face higher dropout risks due to additional responsibilities.
- Parents' education and occupation: Reflect academic support at home.
- Displaced or international status: Indicates a higher dropout due to adaptation challenges.
- Educational special needs: May need additional support from the institution.

- Behavioral and Engagement Features

- Application mode and application order: Capture motivation and decision-making.

- Daytime or evening attendance: Evening students are more likely to juggle work and family responsibilities, raising dropout risk.

Less Useful Features

- Marital Status. This has a weak correlation and a highly imbalanced distribution, adding noise without predictive value.
- Nationality: There are too many categories with few samples, creating sparsity and bias.
- Parents' Occupation: This is vague, inconsistent, and poorly linked to outcomes.
- Macroeconomic Indicators: These aggregate-level variables are not relevant for individual predictions.

Final Feature Selection for Modeling

Based on correlation strength, domain knowledge, and fairness considerations, we selected the following features:

- Admission grade
- Curricular units (1st & 2nd semester: enrolled, credited, approved, evaluations, grade)
- Tuition fee status
- Scholarship holder
- Age at enrollment
- Educational special needs
- Debtor status
- Displaced or international status
- Application mode and order
- Daytime or evening attendance

Models Used

We implemented three models to compare performance:

- Logistic Regression (baseline)
Pros: interpretable and suitable for multiclass classification.
Cons: limited for non-linear relationships.
- Gradient Boosting Classifier
Pros: This is a sequential ensemble method that corrects the mistakes of earlier models. It offers strong predictive performance.
Cons: It needs careful tuning and is more costly in terms of computation.
- Random Forest Classifier
Pros: ensemble of decision trees, reduces overfitting, and captures non-linear relationships.
Cons: less interpretable than a single tree.

Rationale for Selection :

- Logistic Regression offers a straightforward and clear starting point for multiclass prediction.
- Gradient Boosting raises predictive ability by fixing errors from previous models one after another, making it good at identifying complex patterns.
- Random Forest combines accuracy and generalization, minimizing overfitting and serving as the best choice for final results.

Results and Discussion

Performance Before Corrections:

- Gradient Boosting performed best across most metrics, with a validation accuracy of 0.7816, recall of 0.7816, and F1-score of 0.7721. This shows strong generalization and balanced performance. However, its test accuracy (0.7600) was slightly lower than Random Forest.
- Random Forest achieved the highest test accuracy (0.7684), which was very close to its validation accuracy (0.7706). This indicates consistent generalization and reliability across training and testing phases.
- Logistic Regression lagged behind both ensemble models across all metrics, but it still provides a useful baseline reference due to its simplicity and interpretability.
- Overall, ensemble models (Gradient Boosting and Random Forest) clearly outperform Logistic Regression. Gradient Boosting offers the best overall balance of precision, recall, and F1-score, while Random Forest shows slightly

superior performance in terms of test generalization. Logistic Regression, while easier to interpret, sacrifices predictive performance.

Model	Validation Accuracy	Test Accuracy	Precision	Recall	F1-score
Random Forest	0.7706	0.7684	0.7549	0.7706	0.7503
Gradient Boosting	0.7816	0.7600	0.7710	0.7816	0.7721
Logistic Regression	0.7282	0.7297	0.7082	0.7282	0.7081

Performance After Corrections:

- Gradient Boosting continued to perform the best overall, with the highest validation accuracy (0.8095), precision (0.8124), recall (0.8095), and F1-score (0.8090). However, its test accuracy (0.7521) remained slightly below that of Random Forest, indicating a small gap in generalization.
- Random Forest achieved the strongest test accuracy (0.7462), making it more stable on unseen data. Its precision (0.8051) and F1-score (0.7978) were also competitive, though slightly below Gradient Boosting on validation.
- Logistic Regression, while still the weakest among the three, showed stable performance with validation accuracy (0.7250) and test accuracy (0.7238). It remains a useful baseline due to its interpretability, but lags behind ensemble models in predictive power.

Model	Validation Accuracy	Test Accuracy	Precision	Recall	F1-score
Random Forest	0.7976	0.7462	0.8051	0.7976	0.7978
Gradient Boosting	0.8095	0.7521	0.8124	0.8095	0.8090
Logistic Regression	0.7250	0.7238	0.7267	0.7250	0.7240

Fairness metrics:

1. Random Forest

- Gender → Strong bias, favors unprivileged group (SPD < 0, DI > 1)
- International → Almost fair overall, but recall gap exists (EOD ≠ 0).
Scholarship → Strong bias towards scholarship holders (SPD > 0, DI < 0.8).

2. Gradient Boosting

- Gender → Bias present, favors the unprivileged group.
- International → Nearly balanced predictions, but recall gap remains.
- Scholarship → Consistent bias towards scholarship holders.

3. Logistic Regression

- Gender → High disparity, strongest gender bias among models.
- International → Mild disparity against international students.
- Scholarship → Clear bias towards scholarship holders.

1. Random Forest

Protected Attribute	Statistical Parity Difference	Equal Opportunity Difference	Disparate Impact	Comment
Gender	-0.4103	-0.0811	1.3916	Strong bias - Model favors unprivileged gender group (DI > 1).
International	0.0208	-0.4853	0.9842	Almost fair (SPD ≈ 0, DI ≈ 1) but recall gap exists.
Scholarship	0.6734	0.5098	0.6319	Model strongly favors scholarship holders (SPD > 0, DI < 0.8).

2. Gradient Boosting

Protected Attribute	Statistical Parity Difference	Equal Opportunity Difference	Disparate Impact	Comment
Gender	-0.4045	-0.0670	1.4007	Gender bias: favors unprivileged group.
International	-0.0222	-0.3705	1.0175	Nearly balanced predictions, but recall gap remains.
Scholarship	0.6580	0.3970	0.6294	Bias towards scholarship holders.

3. Logistic Regression

Protected Attribute	Statistical Parity Difference	Equal Opportunity Difference	Disparate Impact	Comment
Gender	-0.4228	-0.2595	1.4228	High gender disparity (large SPD and DI > 1).
International	0.1086	-0.0865	0.9148	Mild disparity against international students.
Scholarship	0.6487	0.5096	0.6337	Strong bias towards scholarship holders.

There was a trade-off between accuracy and fairness. The final corrected model is slightly less accurate but significantly fairer.

Contributions

Parts	Manish(in percentage) Roll no - 251330604	Aakanksha(in percentage) Roll no - 210006
Data Collection and Preprocessing	50	50
Data Exploration	50	50
SMOTE	70	30
Feature Selection (Pearson Coefficient)	60	40
Model Selection	70	30
Model Evaluation	60	40
Bias Evaluation	40	60
Report generation	20	80

Conclusion

- Academic performance indicators, including grades, course completion, and evaluations, are the strongest predictors of dropout and success.
- Sensitive features like nationality and marital status have low predictive power but can introduce unfair bias.
- Removing these features and rebalancing classes made the model more equitable, even with a slight drop in accuracy.
- After handling imbalance and bias, both ensemble models (Gradient Boosting and Random Forest) demonstrated stronger and fairer performance, with Gradient Boosting excelling in overall balanced metrics and Random Forest generalizing slightly better on test data. Logistic Regression remains useful for interpretability but is less competitive in performance.

